Labor Supply Effects of Universal Licensing Recognition Laws

Tyler Anderson

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Abstract

Occupational licensing creates barriers to entry and mobility for many occupations in the United States. These barriers create market inefficiencies and reduce welfare, as well as potentially deterring people from entering occupations they would otherwise join. In this paper, I study whether Universal Licensing Recognition laws that reduce barriers to mobility affect the number of people entering licensed professions. I find that the effects of these laws are ambiguous and centered around zero.

1 Intro

Occupational licenses, such as the licenses required for electritions, teachers, and cosmetologists, among others, are typically granted at the state level. When someone with one of these licenses chooses to move states, then getting a new license is often costly and time-consuming. 26 states have now passed some form of universal licensing recognition (ULR).These laws greatly reduce the costs of getting licensed for people

with out-of-state licenses. They allow people with out-of-state occupational licenses to have their licenses recognized by filling out a form (and sometimes paying a small fee) instead of having to meet all of the license requirements of the new state.

In this paper, I study the effect of ULR laws on the decision to enter a licensed profession. I construct an annual state panel data set covering the years 2011-2022. During this time 19 ULR laws go into effect and none of them are repealed. Using data from the American Community Survey and the IRS, I construct a measure of the increase in license portability. The outcome of interest is the percentage of individuals in a state with licenses that are likely to be affected by ULR laws. I restrict my sample to people ages 19-35 and to people living in the same state they were born in to focus on new entrants into licensed professions. I then use two-way fixed effects and staggered roll-out difference-in-differences methods to estimate the effects of this increase in license portability on the number of people in licensed professions. I find that the effects of these laws are ambiguous and my estimates are centered around zero. I find that the effects of these laws are bounded by an increase or decrease of roughly 15%.

Occupational licensing has been steadily increasing in the United States for decades. It is estimated that less that 5% of the U.S. workforce had an occupational license in the 1950's and that 22% of the workforce has a license as of 2024 (Kleiner and Krueger, 2013, Deyo, 2022). Previous literature on occupational licensing shows that licenses increase wages for people in licensed professions. However, they also create barriers to entry that reduce mobility and decrease consumer welfare by increasing prices. This paper contributes to this literature by examining whether reducing costs of interstate migration due to licensing affects people's decision to enter licensed professions.

ULR laws are generally written to apply to as many licenses as possible. Although a small number of statutes specifically list the professions that they apply to, the

large majority list requirements for a license to be recognized without any reference to specific professions. Common requirements are that the licensed individual has a valid license in another state, they are a legal resident of the United States, and that disciplinary actions have not been taken against them recently by the licensing board of their home state. A minority of states with ULR laws also require that the new applicants establish residency in the state that they are applying for a license in, have met licensing requirements that are substantially equivalent to the requirements of the state license they are applying for, or both.

Previous research shows that occupational licenses create barriers to entry that may lead to inefficiencies in the labor market. Kleiner and Soltas (2023) show that occupational licensing reduces employment and welfare. The reductions to employment are caused by the barriers to entry created by licenses, which prevent people form entering licensing professions who, for example, can't afford to take a licensing exam or have a criminal record. Other studies have found that occupational licensing increases wages for licensed professions (Kleiner and Krueger, 2013, Kleiner and Soltas, 2023). There is also strong evidence that occupational licensing reduces worker mobility (Johnson and Kleiner, 2020, Jackson, 2023). This paper contributes to this literature by studying whether lowering the barriers to mobility increases the number of people who enter licensed professions. I show that while the effects are ambiguous, they are centered around 0, which provides evidence that future costs of moving associated with being in a licensed profession is not seen as a barrier to entry for people deciding to enter a licensed profession.

ULR laws may theoretically reduce the barriers to entry created by licensing and therefore the inefficiencies in the labor market. Previous studies of ULR laws have shown that they increase mobility (Deyo and Plemmons, 2022, Bae and Timmons, 2023) and that they decrease wages and increase employment for some people with

these licenses (Bae and Timmons, 2023). However, the effect of these laws on the overall supply of licensed professionals and the decision to enter a licensed profession is ambiguous. ULR laws should increase the value of affected professional licenses by allowing people to use those licenses in more places. Reducing barriers to entry should also increase the supply of workers and decrease the equilibrium wages, thereby reducing the value of the licenses. Bae and Timmons (2023) provide empirical evidence that ULR laws decrease wages and employment for licensed professionals in states with ULR laws. They also show that these laws increase mobility for licensed professionals in all states. However, their study does not address whether wages increase in the states without ULR laws that presumably saw the supply of licensed professionals decrease. It is unclear which of these effects is stronger for people considering entering a licensed profession. This is the first study to examine the effects of ULR laws on the decision to enter a licensed profession. I find that the effect of ULR laws on the number of people in licensed professions is centered around zero and is bounded by an increase or decrease of roughly 15%.

2 Methodology

2.1 Data

I construct an annual state panel data set covering the years 2011-2022. To construct the treatment variable, I use publicly available Internal Revenue Service tax-filer data on state-to-state migration. I use data from the American Community Survey (ACS) from the years 2011-2022 (accessed through IPUMS) for the outcome variable. Information on which states have ULR laws and when they were passed comes from "Survey of Universal Licensing Reforms in the United States" (Deyo, 2022). I also

draw on the work of Bae and Timmons (2023) to identify ACS occupation codes that correspond to professions that require licenses that are relevant to this analysis.

The outcome variable, y_{st}, is constructed using ACS data and is a percentage given by:

$$y_{st} = \frac{\text{\# of licensed people age 19-35}}{\text{\# of people age 19-35}} \cdot 100$$

among people in state s that were born in state s in year t. The sample is limited to residents of the state they were born in to avoid capturing changes in the number of licensed individuals in a state that are caused by people moving after the passage of a ULR law. The age range is restricted to the ages that people are most likely to acquire a professional license to avoid capturing people that are re-entering a profession after the passage of a ULR law. As a robustness check, I use alternative age restrictions in section 4.

The key independent variable is constructed from IRS tax-filer data and data from the Knee Institute on the passage of state ULR laws. This treatment variable, τ, is given by:

$$\tau_{st} = \sum_{d \neq s} \frac{m_{sd}}{m_s} \cdot has_ulr_{dt}$$

where m_{sd} is the number of migrants moving from state s to state d in 2015, m_s is the total number of people that moved from state s to a different state in 2015, and has_ulr_{dt} is an indicator variable equal to 1 if state d has passed a ULR law in year t and 0 otherwise.

The sample includes all 50 U.S. states and Washington, D.C., so this sum includes 50 terms. As an example, consider someone who lives in Colorado in 2018. In 2018, Nevada and New Mexico have passed ULR laws, and about 3% of people from Colorado who moved to a different state moved to New Mexico and about 3% moved to Nevada. The treatment variable for Colorado in 2018 is then about 6%.

Much of my analysis requires a binary treatment variable. For these specifications, I define the variable $treated_{st}$ to be equal to 1 if $\tau_{st} \geq 0.2$ and 0 otherwise. As a robustness check, I use alternative cutoff values in section 4.

Table 1 provides a list of the occupations and the corresponding ACS codes that are considered 'licensed' in the construction of the outcome variable.

Occupation Names
Cosmetologist/barber
Real estate broker/sales agent
Electrician
Manicurist/esthetician
Dental hygienist
Dentist
Massage therapist
Real estate appraiser/assessor
Veterinarian
Funeral director
Podiatrist

Table 1: List of occupations and the corresponding ACS codes included in the sample

This is not a complete list of occupations that are affected by ULR laws. My outcome variable includes professions that are both likely to be affected by ULR laws and that are identifiable by ACS codes. Physicians, for example, are not considered licensed in the outcome variable because there is a pre-existing interstate licensure compact that governs how they become licensed if they choose to practice in a different state. Similarly, there are professions such as waste water workers and polygraph operators that are licensed and covered by most ULR laws that are not identifiable by ACS code and therefore not considered licensed here. There are also professions that are only licensed in one state or a small number of states. As an example, boxing and wrestling promoters are licensed in Virginia but not in other states. These professions

are not considered licensed in this analysis because they are unaffected by ULR laws. The mean proportion of licensed individuals in my sample is 1.8%.

2.2 Estimation

I use two-fixed effects to identify the effect of increased license portability on the percentage of licensed individuals in a state. The estimating equation is:

$$y_{st} = \beta_1 \tau_{st} + \beta_2 has_ulr_{st} + \delta_s + \gamma_t + \varepsilon_{it}$$

The variables y_{st} and τ_{st} are respectively the outcome and treatment variables defined above in section 2.1. I control for the effects of increased competition due to the passage of a ULR law with the variable has_ulr_{st}, which is a binary treatment variable equal to 1 if state s has passed a ULR in year t. State fixed effects are given by δ_s and are included to account for state-specific differences in occupational licensing. Year fixed effects are given by γ_t and are included to account for the upward trend of licensing rates over the time period in the panel across all states.

The coefficient of interest is β_1, which gives the percentage point change in licensed professionals when a state goes from 0% license portability to 100% license portability. This would require every state (except one) to pass a ULR law and is not an outcome that is observed in the data.

Two-fixed fixed effects models have been shown to potentially biased in staggered adoption settings Goodman-Bacon, 2021. As a robustness check, I also use Callaway and Sant'anna difference-in-differences. I run a series of year-by-year and cohort-by-cohort regressions given by:

$$y_{st} = \beta_1 (treated_s \times after_t) + \beta_2 treated_s + \beta_3 after_t + \beta_4 has_ulr_{st} + \varepsilon_{st}$$

This gives a total of 33 regressions, with three cohorts and eleven regressions per cohort. Here, $after_t$ is an indicator variable equal to 1 if t is after the cohort became treated and $treated_s$ is the binary treatment variable defined above that is equal to 1 if $\tau_s \geq 0.2$ in the after period. The variable has_ulr_{st} is an indicator variable that is 1 if state s has passed a ULR law in year t and 0 otherwise. This indicator controls for the effects of increased competition as a result of a ULR law being passed. The average treatment effect on the treated (ATT) is then given by a weighted average of the estimates of β_1.

For comparison with the Callaway and Sant'Anna results, I also use two-way fixed effects with the binary treatment variable, $treated_{st}$, that is equal to 1 if $\tau_{st} \geq 0.2$. The estimating equation with the binary treatment variable is given by:

$$y_{st} = \beta_1 treated_{st} + \beta_2 has_ulr_{st} + \delta_s + \gamma_t + \varepsilon_{it}$$

The first identifying assumption underlying the two-way fixed effects identification is a parallel trends assumption. Specifically, interpreting β_1 as a causal effect of ULR laws requires assuming that the proportion of licensed individuals in highly treated states would have followed similar trends as the proportion of people in barely treated states had these ULR laws not been passed. The second identifying assumption is homogeneous treatment effects. The homogenous treatment effects assumption may not be realistic in the context. I use the Callaway and Sant'Anna DiD design to address concerns over heterogeneous treatment effects. The identifying variation in the data comes from states passing ULR laws over time. As more states pass ULR laws, the value of τ_{st} increases for every state, but by different amounts depending on interstate migration patterns.

It is possible that states pass these laws endogenously in response to labor market

conditions. For example, it could be the case that a state only passes a ULR law when they are experiencing a labor market shortage, and that this shortage then drives up wages and this wage increase causes people to enter the profession while the ULR law has no effect. There are two reasons to believe that this is not a significant concern in this specification. First, my treatment variable captures the effect of ULR laws on states other than the ones that passed them. Therefore, the value of the treatment variable for a state is unrelated to the labor market conditions in that state. Second, these laws were passed over a period of five years across a wide variety of states and they cover a wide variety of professions. It is unlikely that geographically and politically diverse states experienced similar shocks over the course of 2016-2021 and passed ULR laws in response to them. Furthermore, it is unlikely that such shocks would affect all professions in the sample similarly.

3 Results

Table 2 presents the results of both two-way fixed effects and Callaway and Sant'anna staggered-adoption difference-in-differences regressions. Columns 1 and 2 show the results of two-way fixed effects regressions with τ, the continuous treatment variable defined above, as the independent variable. Columns 3 and 4 show the two-way fixed regressions with a binary treatment variable equal to 1 if $\tau \geq 0.2$. Columns 5 and 6 show the Callaway and Sant'anna staggered adoption difference-in-differences regression. Columns 2, 4, and 6 show the regressions controlling for whether state s has passed a ULR law in year t. All of the regression specifications show a positive estimate, and they are all insignificant at the 10% level.

The coefficients on τ can be interpreted as the effect of going from no license portability to perfect license portability on the percentage of licensed people in a

Table 2

	(1) TWFE	(2) TWFE	(3) TWFE	(4) TWFE	(5) CSDiD	(6) CSDiD
τ	0.110	0.0842				
	(0.368)	(0.329)				
has_ulr		0.0205		0.0205		
		(0.0744)		(0.0772)		
Binary ATT			0.0321	0.0293	0.00986	0.00769
			(0.0711)	(0.0676)	(0.126)	(0.127)
Mean	1.8207	1.8207	1.8207	1.8207	1.8207	1.8207
	(0.0175)	(0.0175)	(0.0175)	(0.0175)	(0.0175)	(0.0175)
ATT/Mean	0.0604	0.0462	0.0176	0.0161	0.0054	0.0042
Observations	612	612	612	612	612	612

Dependent variable is percent of licensed people adults in a state. Columns 1 and 2 report TWFE with a continuous treatment variable. Columns 3 and 4 report TWFE with binary treatment variable equal to 1 if $\tau \geq 0.2$. Columns 5 and 6 report Callaway and Sant'anna staggered-adoption differences-in-differences with the same binary treatment variable. Mean is the mean of the dependent variable. Standard errors clustered at the state level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

state. This is not something that occurs in the sample, so comparison with the other coefficients in the table requires multiplying by the difference in the average value of τ for the treated group minus the average value of τ for the untreated group, which is approximately 0.264. Performing this calculation with the coefficient in column 2 shows that the comparable value for the coefficient of interest is 0.0022. This shows that a comparable increase in license portability leads to a 0.002 percentage point (or 0.1 percent) increase in the number of people in licensed professions.

The coefficients on Binary ATT (for binary average treatment effect on the treated) show the effect of going from less than twenty percent license portability to more than twenty percent license portability. Column 6 shows Callaway and Sant'anna difference-in-differences with a control for whether state s has passed a ULR law in year t. The point estimate shows that a state going from less than 20% license porta-

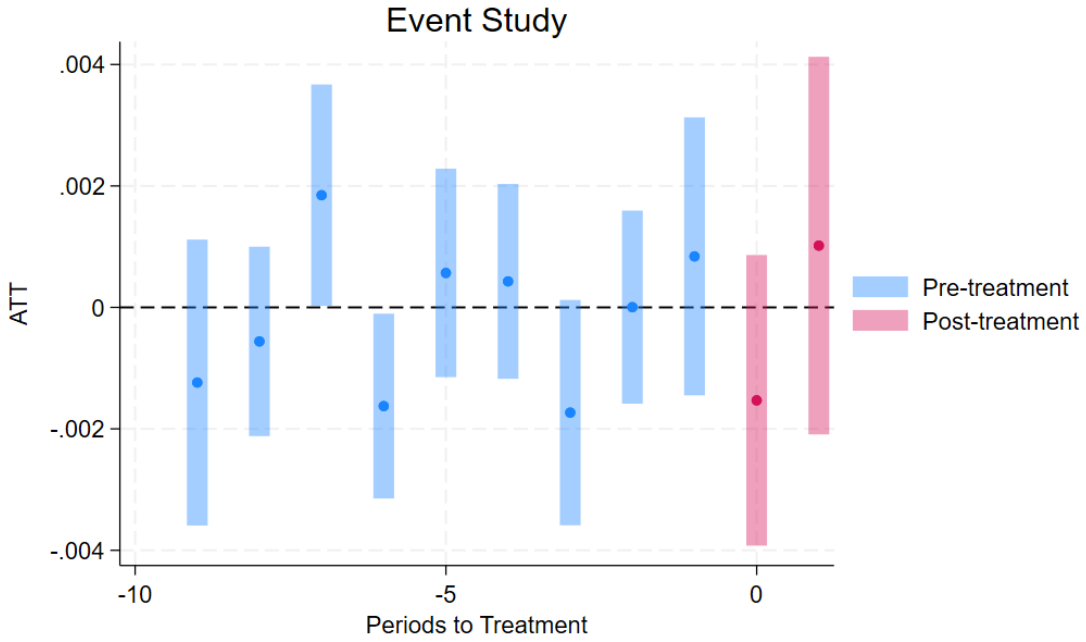


Figure 1

bility to more than 20% license portability causes a 0.008 percentage point (or 0.4%) increase in the percentage of licensed people in a state. The 95% confidence interval of this estimate is -0.2412 to 0.2566, so the effect is bounded between a 13% decrease and a 14% increase.

My analysis relies on a parallel trends assumption between treated and untreated states. To check for evidence of unequal pre-trends, I run an event study using the binary treatment variable as the independent variable and the percentage of licensed individuals (as defined earlier) as the dependent variable. The results of the event study are shown in Figure 1.

The event study figure shows no evidence of pre-trends in the data, but also does not show evidence of a clear change in the trend after treatment. The Callaway and Sant'anna event study shown in Figure 1 as well as the standard event study shown in Figure A.1 provide evidence that there may be an upward trend in the effect of

treatment over time.

Theoretically, it seems likely that the effect of treatment would increase over time for two reasons. First, it may take time for people to become aware of the passage of ULR laws in other states and the increased portability of professional licenses. Second, it takes anywhere between several months and two years for someone who decides to enter a licensed profession to become trained and licensed. It then takes more time for a newly licensed to be surveyed and show up in the data.

4 Robustness

Table 3

	Cutoff values of τ			
Age Range	0.15	0.20	0.25	0.30
19-35	0.145	0.008	-0.0188	0.035
	(0.130)	(0.127)	(0.168)	(0.269)
None	-0.006	-0.029	-0.001	-0.003
	(0.049)	(0.035)	(0.046)	(0.0667)
21-26	0.068	-0.141	0.009	0.216
	(0.146)	(0.158)	(0.157)	(0.229)

Estimates shown are Callway and Sant'anna ATTs. Columns show cutoff values for τ used to determine treatment status. Rows show age range restrictions used for the sample. Standard errors at the state level shown in parenthesis. *** p<0.01, ** p<0.05, * p<0.1

In order to use staggered adoption difference-in-differences methods, the continuous treatment variable, τ, had to be discretized. In the earlier analysis, I chose a cutoff of $\tau = 0.2$, however other choices of a cutoff are reasonable. In table 3, I show the Callaway and Sant'anna ATTs for the same regression as was shown in column 6 of table 2 with cutoffs of 0.15, 0.25, and 0.3. Additionally, I chose to restrict my sam-

ple to people ages 19-35 to capture new entrants into the profession. This age range is based off of the minimum and maximum numbers of people in licensed professions in the data, but other choices are reasonable here. Table 3 also shows the results of restricting the sample to different age ranges (or not at all).

My main analysis has assumed that all individuals have the same probability of moving. In table 4, I relax this assumption by using a treatment variable given by $\tau_{st} \cdot P_s(\text{moving})$ where $P_s(\text{moving})$ is the probability that an individual in state s moves to a different state in 2015. $P_s(\text{moving})$ is constructed from IRS tax filer data and is the proportion of people in state s in 2015 that moved to a different state in 2015. I use three different values of the new treatment variable as a cutoff and the same age restrictions as table 3.

Table 4

	Cutoff values of $(\tau_{st} \cdot P(\text{moving}))$		
Age Range	0.0001	0.0002	0.0003
19-35	-0.002	-0.002	-0.002
	(0.002)	(0.002)	(0.002)
None	-0.001	-0.001	-.001
	(0.001)	(0.001)	(0.001)
21-26	-.002	-.002	-.002
	(0.002)	(0.002)	(0.002)

Estimates shown are Callway and Sant'anna ATTs. Columns show cutoff values for $\tau_{st} \cdot P(\text{moving})$ used to determine treatment status. Rows show age range restrictions used for the sample. Standard errors at the state level shown in parenthesis. *** p<0.01, ** p<0.05, * p<0.1

These changes result in estimates that are similar to the previous analysis, although with smaller confidence intervals. This specification provides evidence that the true treatment effect is very close to 0.

5 Conclusion

This paper shows that the effect of universal licensing recognition laws on the number of people is ambiguous, but centered around zero. Additionally, I show that the effect is at most a fifteen percent increase or decrease in the proportion of licensed people in a state. The implication for policy makers is that these laws should not be used as a tool to increase (or decrease) the number of people in a licensed profession. Further research is warranted on the effects of these laws on prices and quality of services, as well as whether these laws change the composition of people that decide to enter licensed professions.

Appendix A

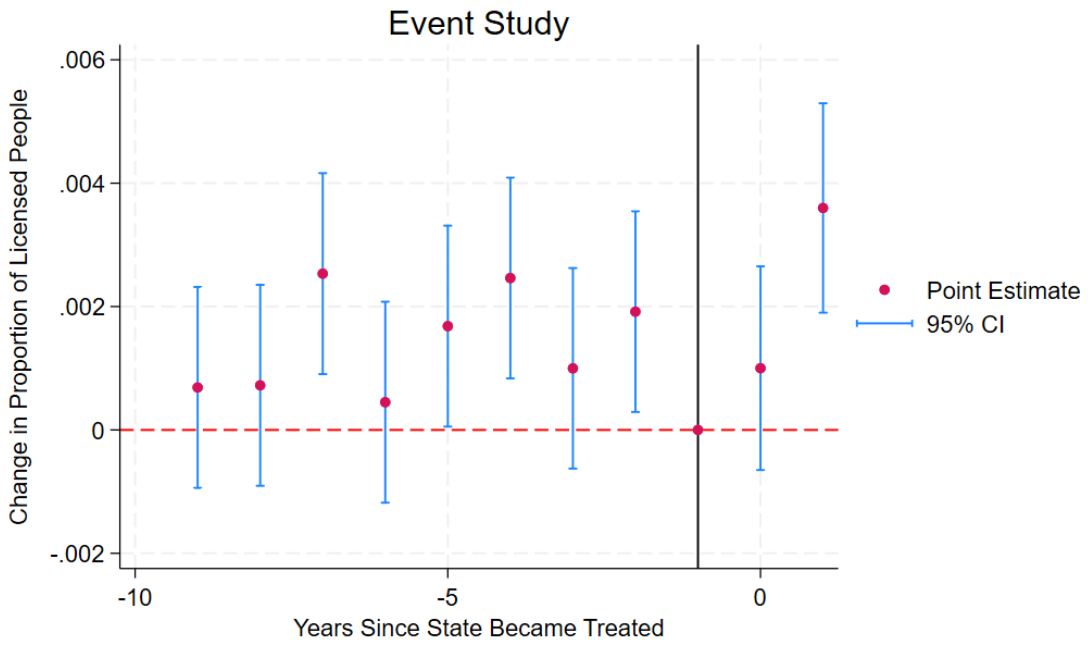


Figure A.1: Standard event study design.

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